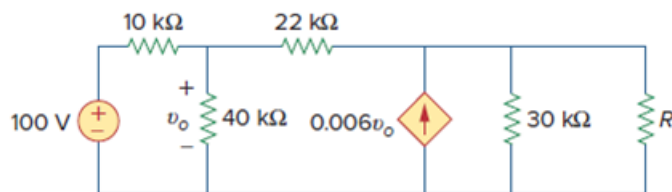


Problem

Find the maximum power transferred to resistor R in the circuit of Fig. 4.135.

Figure 4.135:



Step-by-step solution

Step 1 of 8

Refer to Figure 4.135 in the textbook.

Replace the 100 V voltage source with short circuit and apply 1 V test source in place of the resistor R to calculate the Thevenin resistance, as shown in Figure 1.

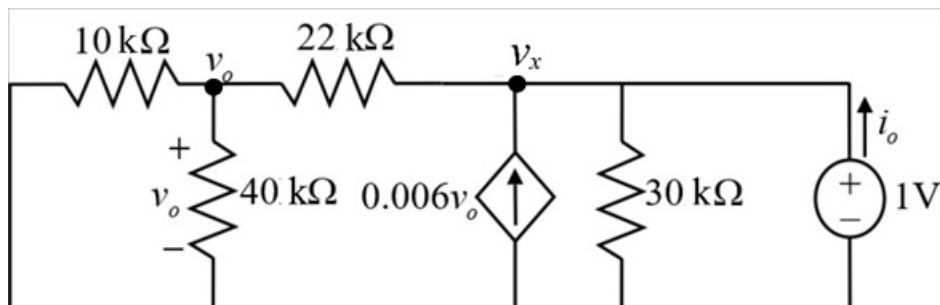


Figure 1

Step 2 of 8

Apply Kirchhoff's current law at node v_o .

$$\frac{v_o}{10 \times 10^3} + \frac{v_o}{40 \times 10^3} + \frac{v_o - v_x}{22 \times 10^3} = 0$$

Since $v_x = 1\text{ V}$ because the $0.006v_o$ current source is parallel to 1 V current source, therefore, substitute 1 V for v_x .

$$\begin{aligned} \frac{v_o}{10 \times 10^3} + \frac{v_o}{40 \times 10^3} + \frac{v_o - 1}{22 \times 10^3} &= 0 \\ \frac{44v_o + 11v_o + 20v_o - 20}{440 \times 10^3} &= 0 \end{aligned}$$

$$75v_o = 20$$

$$v_o = \frac{4}{15}\text{ V}$$

Step 3 of 8

Apply Kirchhoff's current law at node v_x .

$$\frac{v_x - v_o}{22 \times 10^3} - 0.006v_o + \frac{v_x}{30 \times 10^3} - i_o = 0$$

$$v_x \left[\frac{1}{22} + \frac{1}{30} \right] - v_o \left[6 + \frac{1}{22} \right] + 1000i_o = 0$$

$$\frac{v_x [15 + 11] - v_o [1980 + 15] + 33 \times 10^4 i_o}{330} = 0$$

$$26v_x - 1995v_o + 33 \times 10^4 i_o = 0 \dots (1)$$

Since $v_x = 1 \text{ V}$ because the $0.006v_o$ current source is parallel to 1 V current source,

therefore, substitute $\frac{4}{15}$ for v_o and 1 V for v_x .

$$26(1) - 1995 \left(\frac{4}{15} \right) + 33 \times 10^4 i_o = 0$$

$$\frac{390 - 7980 + (495 \times 10^4 i_o)}{15} = 0$$

$$495 \times 10^4 i_o = 7590$$

$$i_o = \frac{7590}{495 \times 10^4}$$

$$= 1.533 \text{ mA}$$

Calculate the Thevenin resistance.

$$R_{Th} = \frac{1}{i_o}$$

Substitute 1.533 mA for i_o .

$$R_{Th} = \frac{1}{1.533 \times 10^{-3}}$$

$$= 0.652 \text{ k}\Omega$$

Step 4 of 8

Redraw the circuit as shown in Figure 2 and calculate the Thevenin voltage.

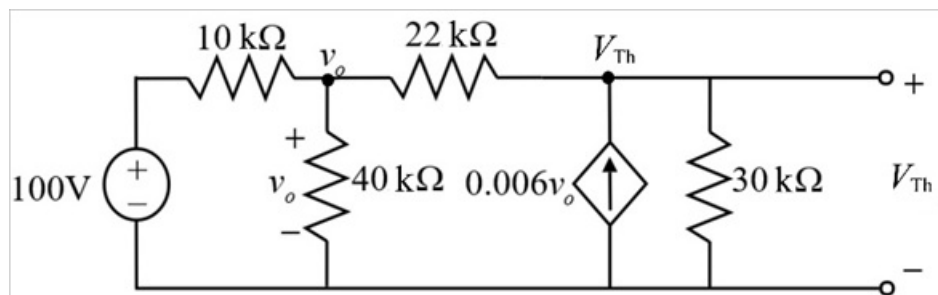


Figure 2

Step 5 of 8

Apply the nodal analysis at the node v_o .

$$\frac{v_o - 100}{10 \times 10^3} + \frac{v_o}{40 \times 10^3} + \frac{v_o - V_{Th}}{22 \times 10^3} = 0$$

$$\frac{v_o}{10} + \frac{v_o}{40} + \frac{v_o - V_{Th}}{22} = 10$$

$$\frac{44v_o + 11v_o + 20v_o - 20V_{Th}}{440} = 10$$

$$75v_o - 20V_{Th} = 4400$$

$$v_o = 0.2667V_{Th} + 58.67 \dots\dots (2)$$

Step 6 of 8

Apply Kirchhoff's current law at node V_{Th} .

$$\frac{V_{Th} - v_o}{22 \times 10^3} - 0.006v_o + \frac{V_{Th}}{30 \times 10^3} = 0$$

$$\frac{V_{Th}}{10^3} \left(\frac{1}{22} + \frac{1}{30} \right) = v_o \left(0.006 + \frac{1}{22 \times 10^3} \right)$$

$$\frac{V_{Th}}{10^3} \left(\frac{15 + 11}{330} \right) = v_o \left(\frac{132 + 1}{22 \times 10^3} \right)$$

$$V_{Th} = \left(\frac{330}{26} \right) \left(\frac{133}{22} \right) v_o$$

$$= 76.73v_o$$

Substitute $(0.2667V_{Th} + 58.67)$ for v_o .

$$V_{Th} = 76.73(0.2667V_{Th} + 58.67)$$

$$V_{Th}(1 - 20.46) = 4501.74$$

$$V_{Th} = -231.33 \text{ V}$$

Step 7 of 8

The Thevenin equivalent circuit is the combination of $V_{Th} = -231.33 \text{ V}$ voltage source in series with $R_{Th} = 0.652 \text{ k}\Omega$ Thevenin resistance and resistor R , as shown in Figure 3.

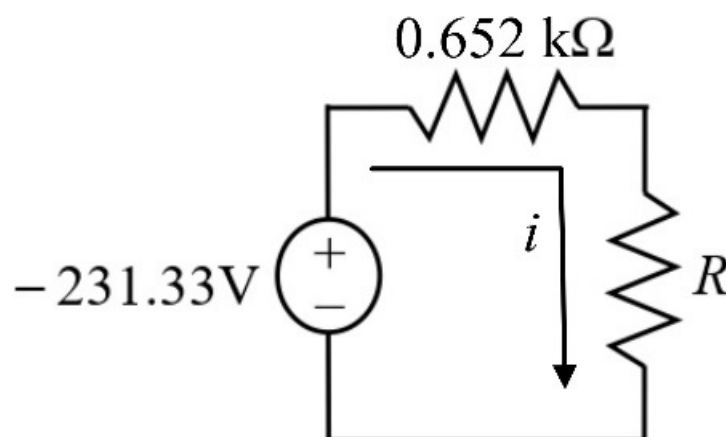


Figure 3

Step 8 of 8

The condition for the maximum power transferred to the load is as follows:

$$R = R_{Th}$$

Substitute $0.652 \text{ k}\Omega$ for R_{Th} .

$$R = 0.654 \text{ k}\Omega$$

Calculate the maximum power to the load.

$$\begin{aligned} p_{\max} &= i^2 R \\ &= \left(\frac{V_{Th}}{R + R_{Th}} \right)^2 (R) \end{aligned}$$

Substitute -231.33 V for V_{Th} , $0.652 \text{ k}\Omega$ for R_{Th} , and $0.652 \text{ k}\Omega$ for R .

$$\begin{aligned} p_{\max} &= \left(\frac{-231.33}{(0.652 + 0.652) \times 10^3} \right)^2 (0.652 \times 10^3) \\ &= 20.52 \text{ W} \end{aligned}$$

Therefore, the maximum power transferred to the load is $\boxed{20.52 \text{ W}}$.